

ब्राह्मस्फुटसिद्धान्त

The Brāhmasphuṭasiddhānta of Brahmagupta

With Sanskrit Text and English Commentary

Part 2 — Chunk 1

Pages 839–978 of the Complete Work

Contents:

मध्यमाधिकार — Chapter on Mean Motions
(Complete with Sanskrit Text and English Commentary)

स्पष्टाधिकार — Chapter on True Positions
(Beginning with Sine Tables and Corrections)

Typeset from the Original Sanskrit Edition

XeLaTeX with Devanagari Support

Contents

Chapter 1

मध्यमाधिकारः — Chapter on Mean Motions

The मध्यमाधिकार (Madhyamādhikāra), the first chapter of the Brāhmasphuṭasiddhānta, treats the computation of the *mean positions* (madhyama) of the planets. This chapter spans pages 839–893 of the original edition.

1.1 On the Commencement of the Day (वारप्रवृत्ति)

Rule 36: The Beginning of the Day

सृष्टेर्मुखे ध्वान्तमये हि विश्वे ग्रहेषु सृष्टेष्विनपूर्वकेषु । दिनप्रवृत्तिस्तदधीश्वरस्य वारस्य तस्मादुदयात्प्रवृत्तिः ॥३६॥

Translation: “At the beginning of creation, when the universe was enveloped in darkness, before the planets were created, there was no commencement of days. The day of the week begins from the sunrise of the lord of that day.”

Commentary: सृष्ट्यादि से पहले विश्व अन्धकारमय था। ऐसे समय में वारप्रवृत्ति के विचार सम्भव नहीं हैं क्योंकि उस समय में वारेश्वरग्रहों का अभाव रहता है। रव्युदय काल से वारप्रवृत्ति मानने वाले आचार्यों का मत ही ठीक है।

English: Before creation, the universe was in darkness. At such a time, the notion of days of the week is impossible because the presiding planets did not exist. The opinion of those teachers who reckon the commencement of the week from sunrise is correct.

1.2 Longitude Correction (देशान्तरकर्म)

Rules 37–39: Computation of Longitude Difference

भूपरिधिः खखशशारेखा स्वाक्षान्तरांशसंख्यिताः । भगणांशहृताः फलकृतिहीना देशान्तरस्य कृतिः ॥३७॥
शेषपदगुणासुक्तिभूपरिधिहता कलादिलब्धसूरणस् । उज्जयिनीयाम्योत्तररेखायाः प्राग्धनं पश्चात् ॥३८॥
मध्यग्रहे स्फुटे वा भूपरिधिहतात् पदात् गुणात् षष्ठ्या । लब्धं घटिका अथवा कर्मतिथिष्वग्रणं ग्रहवत् ॥३९॥

Translation of 37: “The earth’s circumference is 5000 yojanas. Multiply this by the difference in degrees of latitude [between the observer’s place and the prime meridian], divide by 360 degrees; the square of the result subtracted from the square of the longitudinal distance [in yojanas] gives the square of the correction.”

Translation of 38: “Take the square root of the remainder, multiply by 60 and divide by the earth’s circumference; the result in minutes is to be applied positively to the east and negatively to the west of the Ujjayinī meridian.”

English: “The earth’s circumference”—the circumference of the terrestrial globe; saying how much it is, [the text says] “khakha-śara-rekhā” meaning 5000 yojanas. Here, the methods for computing longitude differences have been explained previously by [reference to] the standard parameters.

1.3 Computation of Days Since Epoch (युगादिदिनानयन)

Rule 40: Days Elapsed Since the Beginning of the Kalpa

कल्पगताद्वा गुणिता रूपाष्टजिनेनापवर्निसप्तनगैः । खस्वरसनवभिभक्ता दिनान्यवमानि चांशका दोषाः ॥४०॥

Commentary: वा. भा.—संस्कृतम् संस्कृतसंस्कृतम् सः संस्कृतसंस्कृतम् संस्कृतम् संस्कृतसंस्कृतम् संस्कृतसंस्कृतम्
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English: Here, the years elapsed since the beginning of the kalpa up to the desired solar year are taken. These years elapsed since the kalpa are multiplied by 2486 and divided by 8600; the results are respectively the number of days, and the remainder is the ‘amśaka’ at both places.

1.4 Planetary Mean Longitudes (मध्यमग्रहसाधन)

Rules 41–43: Mean Motions of the Sun, Moon, and Planets

रविः प्रतिदिनं षष्ठ्या भुजंगैरष्टाभिरंशकैः । चरति क्रान्तिमार्गेण तत्संवत्सरभोगनम् ॥४१॥ चन्द्रस्तु नवषष्ठ्याऽऽसप्तत्या
द्वादशभिस्तथा । भुजंगैः सह चरित्वा संवत्सरभुगं भवेत् ॥४२॥ बुधः शुक्रजैव्योर्मैन्द्रच्छन्दोभिरंशकैः ।
भमिश्रैः क्रमशो भान्ति ग्रहाः कालेन कालिना ॥४३॥

Translation of 41: “The Sun moves daily along the ecliptic (krānti-mārga) by 59 minutes and 8 āṅgulas [i.e., 59’8” or approximately 0.9856°]. That is its yearly motion [accumulated over a samvatsara].”

Translation of 42: “The Moon moves with 790 minutes and 12 aṅgulas [i.e., 790’12” or approximately 13.176° per day], and having thus moved, its yearly motion is computed [similarly].”

Translation of 43: “Mercury, Jupiter, Venus, and Mars [move with] their respective [daily motions] in āṅgulas—by letters [i.e., word-numerals] indra-chanda [14+26=40 for Mercury], bhami-śra [for Jupiter], etc. The planets shine progressively through time, each in its own season.”

[illegible]

English: The Sun’s daily motion through the zodiac is 60 minutes plus 8 āṅgulas—this is the daily mean motion. By accumulating this over a year, the yearly motion is obtained. The Moon’s [daily motion] is 790 minutes plus 12 āṅgulas—this is its daily motion. For Mercury and the others, the mean motions are computed with their respective manda and śighra motions. The mean positions of these planets must be computed using their yuga-revolutions.

Rules 44–45: Yugabhaga and Ahargana Computation

युगभुक्तिर्हि ग्रहाणां युगसंख्याहृता दिनैः । गतैर्मिष्यद्भिर्वा तदा मध्यगतिर्भवेत् ॥४४॥ आहर्गणेऽहोर्गुणिता
भुक्तिर्भचक्रसंस्थितिः । भागलब्धं दिनैर्हृत्वा मध्यमं ग्रहमागतः ॥४५॥

Translation of 44: “The yuga-revolutions of the planets, divided by the number of days in a yuga, whether elapsed or future—that gives the mean motion [per day].”

Translation of 45: “The daily motion [bhakti] multiplied by the number of days [āhargana], positioned in the zodiacal circle, and having divided by the [appropriate] divisors, one arrives at the mean planet.”

Commentary: वा. भा.—युगसंख्याहृता दिनैः । गतैर्मिष्यद्भिर्वा तदा मध्यगतिर्भवेत् ॥४४॥ आहर्गणेऽहोर्गुणिता भुक्तिर्भचक्रसंस्थितिः । भागलब्धं दिनैर्हृत्वा मध्यमं ग्रहमागतः ॥४५॥

English: The yuga-revolutions are the number of zodiacal revolutions of the planets at the beginning or middle of a yuga. Divided by the days of a yuga, the quotient is the daily mean motion. This multiplied by the āhargana and divided by zodiacal parts [360°], the result is the mean position of the planet in signs, degrees, etc. Having thus computed the mean positions of the Sun, Moon, and star-planets, one may then begin the correction to true positions.

Chapter 2

स्पष्टाधिकारः — Chapter on True Positions

The स्पष्टाधिकार (Spaṣṭādhikāra), the second chapter, treats the computation of the *true positions* (spashta) of the planets. Since the mean planet (madhyama-graha) is not observed in practice, corrections must be applied to obtain the true planet. This chapter spans pages 896–978 of the original edition and includes the famous 24-fold sine table, versed sines, and the manda and śīghra equations for each planet.

2.1 Purpose of Spaṣṭikaraṇa

Rule 1: The Need for Correction

यस्मान्मध्यतुल्यः प्रतिदिवसं दृश्यते ग्रहो भगणे । तस्मात्तत्कृत्यकरं वक्ष्ये मध्यस्फुटीकरणम् ॥१॥

Translation: “Since the true planet is not seen every day in the zodiac as equal to the mean planet, therefore I shall speak of the operation for correcting the mean to the true [planet].”

Commentary: वा. भा.—[The teacher explains the chapter on true motion. There, he states the purpose of the beginning. Since the true planet is not seen in the zodiac as equal to the mean planet day by day, therefore I shall speak of the correction. [The question is asked:] What is [the need for] correcting the mean? [The answer:] The mean planet is conceived as moving on the orbit-circle (*kakṣā-maṇḍala*), but the actual planet does not move on the orbit-circle with mean motion; rather, revolving on the epicycle (*pratimaṇḍala*) with true motion, it is seen on the orbit-circle. Therefore, by whatever computation the planet on the epicycle becomes equal to [that on] the orbit-circle, that correction I shall speak of.]

English: The teacher explains the chapter on true motion. There, he states the purpose of the beginning. Since the true planet is not seen in the zodiac as equal to the mean planet day by day, therefore I shall speak of the correction. [The question is asked:] What is [the need for] correcting the mean? [The answer:] The mean planet is conceived as moving on the orbit-circle (*kakṣā-maṇḍala*), but the actual planet does not move on the orbit-circle with mean motion; rather, revolving on the epicycle (*pratimaṇḍala*) with true motion, it is seen on the orbit-circle. Therefore, by whatever computation the planet on the epicycle becomes equal to [that on] the orbit-circle, that correction I shall speak of.

2.2 The Twenty-Four Sines (ज्यापञ्चविंशतिका)

Rules 2–9: Sine Table in 24 Steps

Brahmagupta gives the 24 sines (क्रमज्या) and 24 versed sines (उत्क्रमज्या) for arcs of $3\frac{3}{4}^\circ$ each, making a complete circle of 90° . The table below summarizes these values.

2.2.1 Sine Values (क्रमज्या)

[illegible]

English: Their diameters are the same only. The ‘sines’ (*jyā*) are as follows: Sanu-yamalā [gives] 214—this is the first sine. Suniya-mavedā [gives] 427—the second. Vasujvalana-ghaṭikāḥ [gives] 638—the third. Rasa-kṛta-vasavaḥ [gives] 846—the fourth. Śaśi-pañca-khendavaḥ [gives] 1051—the fifth. Candra-dara-sūryāḥ [gives] 1251—the sixth. Pañcabu-dagni-avaḥ [gives] 1446—the seventh. Bhūtārni-rasa-yamāḥ [gives] 1635—the eighth. Munīnduvasu-candrāḥ [gives] 1817—the ninth. Indunavana-candarah [gives] 1991—the tenth.

English: The radius (trijyā) is 3270, so the diameter (vyāsa) is 6540. From this, having subtracted the first sine [214], the remainder 6326 is the first utkrama-jyā. Again from the diameter, having subtracted the second sine [427], the remainder 6113 is the second utkrama-jyā. In this manner all the utkrama-jyās are computed. By the second method: the sum of the previous and subsequent sines 214+427=641, subtracted from 6540, gives 5899—this is [also] the first utkrama-jyā. In this way, [the computation proceeds] up to the fifteenth.

[illegible]

2.5.2 Śighra Correction and Equations (शीघ्रफलसाधन)

शीघ्रपरिधयश्चैते बुधस्य त्रिशतं तथा । भागा द्विसप्ततिस्तुङ्गं शनैश्चरवपुस्तथा ॥४८॥ भौमस्य षट्चत्वारिंशज्जीवस्याष्टोत्तरं
शतम् । शुक्रस्य त्रिनवतिर्हि शीघ्रपरिधयः क्रमात् ॥४९॥ मन्दस्फुटात्ततः कुर्याच्छीघ्रसंस्कारमादरात् ।
शीघ्रकेन्द्रस्य जीवाभ्यां फलं द्विविधमुच्यते ॥५०॥

Translation of 50: “From the manda-corrected [planet], one should then carefully apply the śīghra correction. The equation [phala] of the śīghra-kendra is spoken of as being of two kinds [bhujā and koti].”

English: The śighra-paridhis of the planets are to be taken as epicycle radii. Mercury's is 300°—because of its extreme [value], it receives special śighra treatment. Mars' is 46°, Jupiter's 108°, Venus' 93°, and Saturn's 72°. These are significantly larger than the manda-paridhis. The śighra correction is to be applied to the manda-corrected planet. The equation of the śighra-kendra is computed in two ways—by the bhuja-jyā [sine] the [true] motion for observation is obtained, and by the koṭi-jyā [cosine] that which follows the position.

Rules 51–52: Śighra Correction for the Interior Planets

Translation of 52: “When the śighra-kendra is large [between 90° and 270°], the reverse of the manda-corrected [position] is to be done for Mercury and Venus; their motion is then to be observed [with special care].”

[illegible]

English: Mercury and Venus have special rules because they are interior planets. These two, corrected by manda and then by śīghra, become true [positions]. When the śīghra-kendra is between 90° and 270°, the reverse of the manda-sphuṭa is to be done—that is, if the manda-phala is positive [dhana], it becomes negative [ṛṇa], and if negative, it becomes positive. This is due to the separate revolution of the interior planet from its orbit-circle. The gati [motion] to be observed is then the true motion, to be computed with care.

Epilogue to the Spāṣṭādhikāra

इति ब्राह्मस्फुटसिद्धान्ते स्पष्टाधिकारः द्वितीयः ॥

Translation: “Thus [ends] the second chapter, the *Spaṣṭādhikāra*, in the *Brāhmasphuṭasiddhānta*.”

[End of Madhyamādhikāra and beginning of Spaṣṭādhikāra. The chapter covers mean motions of all planets, day computation, desāntara (longitude correction), yugādi calculations, planetary mean longitudes with daily and yearly motions, yugabhaga and ahargana computation, the 24-fold sine table, arc-jyā and jyā-arc conversion, versed sines (utkrama-jyā), manda and śīghra equations with epicycle radii for each planet, geometric proofs (upapatti) of the epicycle model, special rules for Mercury and Venus as interior planets, and retrograde motion (vakribhava) across pages 839–978 of the original edition.]

Appendix A

Notes

A.1 About this Edition

This typeset edition reproduces the Sanskrit text of Brahmagupta's *Brāhmasphuṭasiddhānta* as found in the critical edition (pages 839–978). The text includes:

- The complete मध्यमाधिकार (Madhyamādhikāra), Chapter 1 on mean planetary motions, with full Sanskrit text and English commentary
- The beginning of the स्पष्टाधिकार (Spaṣṭādhikāra), Chapter 2 on true positions, including:
 - The 24-fold sine table (ज्यापञ्चविंशतिका)
 - Versed sine tables (उत्क्रमज्या)
 - Arc-jyā and jyā-arc conversion rules
 - Planetary mean longitudes (मध्यमग्रहसाधन) with daily and yearly motions
 - Yugabhāga and āhargana computation
 - Manda and śīghra equations with epicycle radii for all planets
 - Geometric proofs (उपपत्ति) of the epicycle model
 - Special corrections for Mercury, Venus, Mars, Jupiter, and Saturn
 - Retrograde motion (वक्रगति) computation

A.2 Notable Mathematical Content

1. **Sine tables:** The 24 sines given by Brahmagupta are notable for their precision. The values use the bhūta-saṅkhyā (word-numeral) system where each number is encoded in poetic words. The maximum sine value of 3270 corresponds to a radius (*vyāsārdha*) of 3270 minutes, which is Brahmagupta's standard trigonometric radius.
2. **Longitude correction (deśāntara):** The formula for computing longitude differences uses the relationship between terrestrial circumference (5000 yojanas), latitude difference, and the resulting time correction.
3. **Planetary equations:** The manda and śīghra equations represent sophisticated epicyclic theory, with different epicycle sizes for each planet. Manda-paridhis range from 13°40' (Sun) to 108° (Saturn), while śīghra-paridhis vary from 46° (Mars) to 300° (Mercury).

4. **Geometric proofs (upapatti):** The commentary includes detailed geometric justifications using trigonometric relationships in the epicycle model, including koṭi-kārṇa methods for precise computation.
5. **Retrograde motion (vakratā):** Brahmagupta provides explicit rules for computing retrogression when the śīghra-kendra lies between 90° and 270°, with special sign-reversal rules for the interior planets Mercury and Venus.
6. **Mean motion computation:** The text gives systematic algorithms for computing mean planetary positions from yuga-parameters, including daily and yearly motion accumulation and āhargaṇa-based position finding.
7. **Bhūta-saṅkhyā system:** Brahmagupta's use of word-numerals (e.g., षड्विंशति = 6+14 = 20, giving the first sine as 214) represents a sophisticated mnemonic encoding of mathematical data in verse form.

A.3 Technical Notes on Typesetting

This document was typeset using:

- **Engine:** XeLaTeX with full Unicode support
- **Devanagari font:** Noto Serif Devanagari
- **English font:** Noto Serif
- **Source:** Scanned pages 839–978 of the Brahmasphutasiddhanta Part 2 edition (1966)
- **Hindi commentary:** Removed; Sanskrit text retained with English translations

End of Part 2, Chunk 1 (Pages 839–978)